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Driving Resilient and Sustainable Marine Operations and Logistics in Asia & India: Aligning Subsea Technologies, Offshore Operations, and Human Capital for the Energy Transition

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Abstract

Asia and India stand at the crossroads of a maritime transformation shaped by climate goals, digital disruption, and shifting geopolitics. This paper, *Driving Resilient and Sustainable Marine Operations and Logistics in Asia & India*, explores how the region can shift from fragmented, carbon-heavy systems to resilient, future-ready engines of energy transition. It highlights challenges like regulatory fragmentation, taxation, ageing assets, climate risks, and skill shortages while showcasing solutions in digitalization, decarbonization, and subsea innovation. Anchored by India's *Sagarmala and Maritime Amritkaal Vision 2047*, the blueprint envisions oceans evolving into hubs of clean energy, connectivity, and sustainable economic security.

Executive Summary

Slogan for Marine trade: "Water's inherent property of buoyancy has saved the world from extinction"

India and Asia today stand at a decisive maritime juncture, influenced by geography, geopolitics, and economic imperatives. As global energy systems shift under the dual forces of climate commitments and rapid digital transformation, the region's marine operations are evolving beyond their traditional role in oil and gas supply. They are increasingly viewed as strategic enablers of the energy transition itself.

This paper outlines how Asia's maritime infrastructure, with India as the central anchor, is preparing for a new era of resilient, sustainable, and technology-driven offshore operations. It draws on developments in subsea engineering, offshore logistics, environmental governance, digital systems, and workforce strategies to frame Asia's response to the challenges of the energy transition.

Key Challenges

The sector continues to face entrenched obstacles: disjointed regulatory frameworks, varied taxation rules, seasonal monsoon disruptions, ageing infrastructure, and workforce skill gaps. These issues reduce efficiency and weaken resilience. For example, mismatched customs and clearance processes across borders delay projects, while complex tax structures inflate transaction costs. Seasonal storms and cyclones regularly disrupt port operations, underlining the urgency of climate-adapted infrastructure and AI-enabled

forecasting. Meanwhile, the absence of advanced training programs hampers the adoption of modern technologies. Overcoming these barriers is essential as energy demand grows and the cargo mix increasingly includes renewable equipment and new fuels.

Digitalization and Decarbonization- Twin Pillars of Transformation

Despite hurdles, two forces are propelling transformation: **digitalization** and **carbon reduction**. Artificial intelligence is already being deployed for vessel scheduling, predictive routing, and streamlined customs, reducing idle time and cutting emissions. Blockchain-based cargo tracking and real-time port platforms are moving the sector toward transparent, paperless operations. In parallel, the adoption of hybrid propulsion systems, cleaner fuels such as LNG, ammonia, and hydrogen, and green infrastructure (like shore-power systems and low-impact breakwaters) signal a decisive move toward sustainability. Ports are also being redesigned to handle renewable cargoes such as offshore wind components and hydrogen storage units.

Subsea Technologies and Offshore Innovation

A core aspect of the paper is aligning subsea & off-shore technologies with these sustainability goals. India & her neighbors are investing in modern subsea infrastructure- from expanded LNG pipelines and subsea power cables to autonomous subsea systems for inspection and maintenance. Notably, the study maps India's recent advances in offshore capabilities, including new LNG import terminals, subsea pipeline networks, offshore support vessel (OSV) construction, and deployment of cutting-edge platforms. Emerging technologies such as gravity-based offshore platforms (for example, gravity-based terminals) and floating storage and regasification units (FSRUs) are enabling faster, more resilient deployment of energy infrastructure. Asian waters are also pioneering autonomous subsea drones (unmanned) & remotely operated vehicles (ROVs) that can perform underwater inspections and repairs, reducing the need to send humans into risky environments. These innovations contribute to safer and more efficient offshore operations, which is crucial as the region explores offshore wind farms, tidal energy projects, & other renewable installations. Indeed, the paper notes that hydrogen infrastructure is becoming a key element of marine logistics: ports are planning hydrogen bunkering facilities & even floating hydrogen storages, while experimental hydrogen-powered vessels and autonomous port service craft are on the horizon. By embracing such subsea and offshore technologies, Asia can ensure that its marine logistics backbone remains resilient and future-ready as energy systems diversify.

Workforce Modernization and Policy Alignment

Technology alone cannot drive this transformation; equally important is investing in people and policy. This paper emphasizes the need for workforce development and upskilling to create a talent pool adept in digital and sustainable marine operations. Currently, South Asia lacks unified standards for maritime education and certification, which hampers labor mobility and the adoption of best practices. To overcome this, the paper advocates new initiatives like a regional Maritime Human Capital Development Fund and collaborative training programs to equip workers with skills in port automation, AI, and subsea engineering. Modernizing the workforce goes hand-in-hand with regulatory harmonization. This study calls for greater alignment of regulations and standards across Asian maritime nations to streamline operations. For example, a Marine Single Window digital system could unify environmental clearances, customs, and shipping permits in one platform to cut red tape for new projects. Regional mutual recognition agreements on crew certifications and safety protocols would likewise allow smoother cooperation and resource sharing. Governments can introduce green financing, tax breaks, & carbon-linked port incentives to encourage ports and ship operators to adopt cleaner technologies. By updating policies, from simplifying marine tax regimes to setting common decarbonization targets; Asian countries can create a more integrated and innovation-friendly maritime ecosystem.

India's Position and Regional Collaboration

India is positioned as the geographic and strategic anchor for Asia's marine energy transition. With a 7,500 km coastline, over 200 ports, and a location bridging the oil-rich Middle East and the energy hungry economies of East Asia, India's role is pivotal. Under the able leadership, domestic initiatives like the *Sagarmala* Programme, a nationwide port modernization and connectivity drive and the Maritime *Amritkaal* Vision 2047, a long-term blueprint for a digitized, sustainable maritime economy underscores India's commitment to maritime revitalization. These programs are not only upgrading India's port infrastructure but also aligning it with future needs such as AI-assisted logistics, autonomous vessels, and offshore renewable energy projects. *Sagarmala's* port network can serve as a backbone for broader regional integration of supply chains. Indeed, India is poised to lead by example; its policies can catalyze region-wide standards and interoperability. Frameworks like ASEAN, BIMSTEC, & the Indian Ocean Rim Association (IORA) should be leveraged for regional maritime collaboration, standardization, and data-sharing. Enhanced cooperation through these bodies could produce, for instance, a coordinated "South Asian Marine Corridor" connecting ports with shared digital platforms and green shipping corridors for LNG, hydrogen, and renewables. By harmonizing policies and jointly investing in cross-border infrastructure, Asian nations can multiply the benefits of individual efforts. The vision is for an Indo-Pacific maritime network where ports from the Middle East, South Asia, and Southeast Asia operate on common sustainability and digital standards, drastically improving efficiency and resilience of energy logistics. Asia's diplomatic and economic leaderships will be critical in aligning these diverse stakeholders and driving a united regional agenda.

A Holistic Blueprint for Asia's Marine Energy Future

In sum, driving resilient and sustainable marine operations and logistics in Asia & India provides a timely, holistic blueprint for transforming the marine logistics sector into a pillar of the clean energy future. It combines diagnostic insight with strategic guidance, making it relevant to policymakers, industry planners, and academic experts alike. This paper not only diagnoses the current gaps in infrastructure, technology, regulation, and workforce but also offers practical solutions and a forward-looking roadmap. From AI-driven port management to autonomous subsea systems, and from hydrogen ready ports to gravity-based offshore platforms, the strategies detailed are ambitious yet grounded in real world case studies and pilot projects. The overarching message is the world, anchored by Asia's proactive initiatives, can redefine its marine sector from a supporting role to a driving force of the 21st-century energy transition. By aligning subsea technologies, offshore operations, and human capital development with the sustainability agenda, the region's maritime logistics infrastructure is steadily being future-proof. This transformation emphasizing digital innovation, decarbonization, regional cooperation, and skilled human capital holds strategic importance not just for Asia's economy, but for global energy security and climate goals. The coming decades will witness Asia's ports and seabed laden with new energy projects, and ensure that the region's marine operations evolve resiliently and sustainably to meet those challenges.

Introduction

As the tides of global energy transition rise, the oceans of Asia are being reimagined; not merely as trade corridors, but as dynamic engines of sustainability, innovation, and geopolitical influence. Ports are no longer just gateways for container traffic but they are becoming nerve centers for decarbonized fuel distribution, offshore renewable support, and digital trade facilitation. And at the crossroads of these transformations stands India and Asia, emerging as a regional anchor in the evolving marine logistics ecosystem in the Indo-Pacific arena.

The maritime infrastructure of Asia has traditionally operated in fragmented silos separated by jurisdictional complexities, varied regulatory regimes, and disparate levels of technological maturity. Yet,

as the region contends with increasingly volatile weather, shifting cargo profiles, and growing demand for clean energy, these silos are giving way to a new vision- interconnected, smart, and climate-resilient marine operations.

Asia's Maritime Awakening

In recent years, countries across the continent have initiated strategic overhauls of their maritime sectors. Indonesia's Pelindo Integration Plan, completed in 2021, unified four state-owned port authorities into a single national entity which resulted in streamlining management and enabling a long-term roadmap for LNG handling, digitalization, and infrastructure harmonization. Vietnam has pursued an ambitious Green Port Initiative under its Ministry of Transport, introducing emission control zones, clean vessel incentives, and shore-power systems in Haiphong and Cai Mep-Thi Vai ports.

Meanwhile, Singapore's Tuas Mega Port, the largest fully automated terminal in the world, is not just a technological marvel, it is also hydrogen-ready and embedded with a smart data layer that integrates with maritime traffic across Southeast Asia. South Korea and Japan are equally invested in the future, with hydrogen-fuel shipping corridors, autonomous cargo vessel testing zones, and integrated port-to-hinterland digital systems forming part of their national maritime plans.

Across these developments, one theme is evident i.e., Asia is accelerating its transformation from a fragmented maritime bloc to a cohesive oceanic network which is digital, decarbonized, and strategically interoperable.

India's Strategic Geography and Maritime Ambition

India's maritime geography has long defined its historical role in regional trade and geopolitics. With access to the Arabian Sea and the Bay of Bengal, proximity to the Strait of Hormuz and Malacca, and a coastline dotted with over 200 ports, India is uniquely positioned to serve both as an energy gateway and as a logistics integrator. The *Sagarmala* Programme, initiated in 2015, laid the groundwork by identifying over 600 port-linked infrastructure projects, aimed at reducing logistics costs and increasing modal share for coastal shipping. More recently, the Maritime *Amritkaal* Vision 2047 has articulated an even broader ambition envisioning green port infrastructure, AI-driven operations, autonomous vessels, & full integration of Indian ports into global value chains.

In this vision, marine logistics becomes not merely an enabler but a pillar of national energy security and regional diplomacy. Ports like Paradip, Kandla, and Tuticorin are evolving into multi-cargo, hydrogen-ready clusters. Inland waterways are being revitalized to link industrial hubs with deep-sea ports. National agencies are collaborating with regional and global bodies from BIMSTEC & IORA to the International Maritime Organization (IMO) to harmonize standards, enhance safety, and share digital marine intelligence.

India's position is further strengthened by its active participation in regional energy diplomacy. The development of Chabahar Port in Iran, agreements with Bangladesh for coastal shipping and transshipment, and trilateral projects with Thailand and Myanmar signal a clear intent to converge regional maritime infrastructure under shared norms for trade, sustainability, and technological adoption.

The Shifting Cargo Profile: Energy Transition at Sea

The nature of cargo itself is undergoing radical change. Whereas the last century was dominated by crude oil, coal & containerized goods, the next few decades will increasingly see ports handling hydrogen cylinders, ammonia, offshore wind turbine components, and floating solar hardware. This shift is not only reshaping yard & berth configurations but demanding new inspection, repair & maintenance (IRM) capabilities.

Subsea technologies are central to this change. From ROVs (Remotely Operated Vehicles) managing deep-water pipelines to autonomous underwater drones inspecting umbilical, Asia's sea-beds are rapidly becoming critical operational zones. India, with proven expertise in subsea pipelaying from the Mumbai

High Basin to the Krishna-Godavari cluster, is now advancing toward more modular, hybrid platforms, integrating gravity-based structures with digital IRM systems powered by artificial intelligence.

At the same time, offshore logistics are adapting to the needs of floating terminals, FPSOs, and deep-water ammonia storage. The rise of Emergency Pipeline Repair Systems (EPRS), digital twin-based EPCI planning, and DP2-enabled crew vessels reflect a sector evolving to support not just efficiency but safety, precision, and resilience in extreme environments.

Technology, Taxation, and the Human Element

Digitalization remains a cornerstone of Asia's maritime transition, though uneven in its implementation. While Singapore & South Korea lead with real-time scheduling, automated cranes, and AI-aided berth planning, many South Asian ports still operate under manual or partially digitized systems. The opportunity lies in scaling digital marine platforms; standardized APIs for customs clearance, predictive weather-routing, and integrated port community systems across borders.

Fiscal reform is essential too. Taxation in marine trade in Asia remains fragmented, with inconsistencies in customs duty on subsea assets, lack of clarity on green fuel incentives, and limited coordination in offshore EPC taxation. A harmonized Blue Marine Taxonomy, aligned with IMO decarbonization goals, could attract sustainable finance & reduce project risks across the region.

And finally, no transformation can succeed without a skilled, future-ready maritime workforce. Today's seafarers and port engineers must be retrained not only in AI and data analytics, but in climate adaptation, offshore renewables & sustainable bunkering. India's proposal for a National Marine Skill Development Program (NMSDP), coupled with mutual recognition agreements under BIMSTEC, could form the basis of a shared human capital strategy for Asia's ports & offshore systems.

A Turning Point in Asia's Maritime Century

As the energy landscape tilts toward offshore wind, hydrogen fuels & subsea digital systems, the maritime sector stands at a turning point. India, with its economic scale, geographic advantage, and policy momentum, is well placed to guide a broader Asian renaissance in marine operations. Regional frameworks are beginning to converge; digital standards are emerging and the cargo of the future is knocking at port gates.

This transformation though complex presents a once-in-a-generation opportunity. If Asia can align its subsea technologies, offshore logistics, and human capital with the imperatives of sustainability and resilience, it will not only secure its own energy future but redefine global maritime norms for decades to come.

Marine Operations & Logistics in India and South Asia- Landscape and Challenges

The Maritime Corridor of South Asia: From Past to Present

South Asia's maritime domain has historically been crucial yet underdeveloped. In the pre-2000 era, most ports in India and its neighbors were government-run, with limited containerization and dated infrastructure leading to long ship turnaround times and congestion. Global trade growth and India's 1990s economic liberalization set the stage for change in the new millennium. Today, roughly 90% of world trade by volume travels by sea, and the Indian Ocean features prominently: over two-thirds of global oil shipments and one-third of bulk cargo (such as coal and iron ore) transit this water. Vital chokepoints like the Strait of Malacca carry about 30% of all global traded goods, while the Hormuz Strait sees ~20% of worldwide oil and liquefied natural gas flow. Positioned astride these routes, South Asia's seas and ports have gained immense strategic importance in global logistics.

This importance is especially evident in energy trade. India is now the world's third-largest oil consumer, importing ~85% of its oil needs (around 4.6–5 million barrels per day), and the region's LNG imports are

rising fast (India alone consumes ~20–25 MTPA of LNG annually). Most of this oil, gas, and bulk cargo is moved by ship, making ports the lifeline of South Asia's energy security and commerce. India's 7,500 km coastline is dotted with 12 major ports and over 200 minor ports, which together handle about 95% of the nation's trade volume. Other South Asian nations like Sri Lanka, Bangladesh, and Pakistan also rely on seaports for the majority of their trade. The region's maritime trade volume has surged post-2000, in step with economic growth and energy demand for example, India's ports now handle over 1.3 billion tons of cargo annually! A figure that has grown steadily in the past two decades. Encouragingly, port performance has started improving: India's average turnaround time for ships is now under 1 day at major ports, and the country ranks 22nd in the world on the Logistics Performance Index (2023) for international shipments, reflecting progress in efficiency. At the same time, nine Indian ports have entered the world's top 100 by volume as of 2023, underscoring the region's growing role.

Yet, despite this growth and global connectivity, South Asia's maritime operations landscape remains fragmented and under-optimized, facing several structural challenges. The following sections examine these key challenges which spans port governance, regulatory bottlenecks, infrastructure gaps, climate vulnerabilities, and workforce limitations while also highlighting emerging trends (like renewable energy and hydrogen) reshaping the sector.

The Emerging Importance of South Asia's Maritime Corridor

South Asia is one of the fastest-growing energy-consuming regions globally, with India alone accounting for over 5 million barrels per day of crude oil imports and over 25 MTPA of LNG imports. The region's vast coastline, dense population centers, and high import dependency position marine logistics as a critical enabler of both economic growth and energy security. As offshore energy activities and energy transition projects grow, ports and marine corridors across India, Sri Lanka, Bangladesh, and the Maldives will bear increased logistical pressure.

However, the existing marine logistics and operations infrastructure remains under-optimized, fragmented, and vulnerable to climate, regulatory, and policy-related risks. This chapter presents the current landscape while identifying key structural challenges.

Fragmented Port Operations

India currently has 12 major and 200+ minor ports, most governed by distinct state and central agencies. The lack of unified logistics planning, duplicative port investments, and insufficient multimodal connectivity results in inefficiencies and prolonged turnaround times. For instance, while ports like *Mundra and Krishnapatnam, India* have modernized significantly, many government-run ports still struggle with manual scheduling, under-capacity berths, and congestion.

In South Asia, this challenge is exacerbated by diplomatic and policy friction. For example, Bangladesh's Mongla port lacks energy cargo handling infrastructure despite being strategically positioned. Sri Lanka's Hambantota Port, though physically advanced, remains underutilized due to lack of regional integration.

Regulatory Overlaps and Inconsistencies

Marine logistics across India and South Asia face regulatory bottlenecks due to overlapping jurisdictions—customs, environment, shipping, DGFT, and port authorities often operate in silos. For example:

- LNG import operations often face delays due to redundant clearances from pollution control boards and state maritime boards.
- Subsea cable laying or pipelay operations face contradictory permissions from environment and seabed authorities.
- Such delays discourage foreign direct investment (FDI) in marine infrastructure, slow EPC schedules, and increase project risk.

Limited Digitalization and Legacy Port Infrastructure

Although private players like Adani Ports have introduced digital port community systems and automated berth allocation, many public ports still rely on manual vessel registration, offline documentation, and non-real-time cargo tracking.

As a result:

- Customs clearance delays can add 2-4 days to ship turnaround.
- Port performance metrics (berth occupancy, yard utilization) are inaccurately reported.
- There is no centralized digital marine operations dashboard across South Asia, leading to inconsistent data sharing and operational opacity.

Monsoon and Climate Vulnerability

The Bay of Bengal and Arabian Sea experience intense monsoon swells, cyclones, and high tides, particularly during Q2 and Q3. This seasonal disruption affects:

- Offshore construction windows for breakwaters and pipelines
- Safe berthing and cargo discharge at ports
- Crew transfer and offshore wind project logistics

Most regional ports lack real-time weather routing systems, AI-enabled forecasting, or automated port shutdown protocols, increasing risk to assets and personnel.

Workforce Skill Gaps

Marine operations are increasingly technology-intensive, demanding skilled professionals in subsea engineering, port automation, and offshore digital systems. However:

- India faces a shortage of certified ROV/AUV pilots, underwater welders, and marine AI programmers.
- Training institutes are few, and curriculum modernization lags behind global standards.
- South Asia lacks a shared workforce certification framework, unlike the EU's STCW model.

Opportunities Amidst the Challenges

Despite these hurdles, India and South Asia have a strategic opportunity to transform marine logistics:

- India's Sagarmala Project can serve as a backbone for regional port integration.
- BIMSTEC and IORA frameworks can harmonize maritime data and policy.
- Private-public collaboration, especially with emerging marine-tech start-ups, can accelerate digitalization.
- Ports can become energy innovation zones, supporting hydrogen/ammonia terminals and offshore renewables logistics.

Strategic Solutions to Marine Logistics Challenges in South Asia

While India and South Asia face several interlinked challenges in the domain of marine logistics-ranging from regulatory fragmentation to climate vulnerability-these issues are not insurmountable. On the contrary, they present a unique opportunity to leapfrog legacy systems and embrace digitally enabled, sustainability-led marine operations. This section provides targeted, solution-oriented recommendations mapped to each challenge previously outlined.

Unified Port Governance and National Marine Coordination Centre. Solution to Fragmented Port Operations (3.2):

Establish a National Marine Logistics Coordination Authority (NMLCA) under the Ministry of Ports, Shipping and Waterways (MoPSW), with a mandate to unify operations across major and minor ports, remove duplicative investments, and create integrated port clusters.

Additionally:

- Introduce Port Performance Benchmarking tools across all ports
- Encourage hub-and-spoke models in coastal shipping
- Digitally connect ports via a National Port Grid for real-time data exchange

Regulatory Harmonization via Marine Single Window System. Solution to Regulatory Overlaps (3.3):

Develop a Marine Single Window (MSW) platform, where environmental, customs, shipping, and state approvals are processed through a single digital interface. This can drastically reduce EPC delays, especially for FSRUs, LNG terminals, and subsea projects.

The MSW should:

- Operate under DG Shipping with API integrations to State Maritime Boards
- Include auto-approval triggers for low-risk operations
- Be built using blockchain-based authentication for traceable clearance

Port Digitalization and Smart Infrastructure Fund. Solution to Digitalization Gaps (3.4):

Create a "Smart Ports Acceleration Fund" to co-finance the upgrade of public ports. Funds could be used to install:

- Automated berth and yard systems
- AI-based vessel arrival forecasts
- Smart Customs Clearance Zones using digital KYC and pre-verified cargo

Introduce the Port Digital Maturity Index (PDMI) to rate port tech-readiness and incentivize investment.

Climate Resilience and Monsoon Management. Solution to Monsoon Disruptions (3.5):

Mandate all ports above a certain throughput to install:

- AI-powered weather-routing platforms
- Automated emergency response plans integrated with national disaster systems
- Flexible floating infrastructure like modular pontoons and offshore transshipment barges

Additionally, introduce marine climate insurance products for EPC contractors to de-risk project schedules.

National Marine Skill Development Program (NMSDP). Solution to Workforce Skill Gaps (3.6):

Launch the National Marine Skill Development Program (NMSDP) in partnership with institutions like IMU, IIT-Madras, and international certifying bodies (ABS, DNV).

Key components:

- Mobile training units for coastal communities

- Tiered certification in port automation, subsea inspection, AI programming
- Funding via CSR mandates from marine PSUs and private port operators

Green Infrastructure Mandates and Fiscal Incentives. Solution to Legacy Infrastructure and Carbon Footprint (3.4 & 3.5):

Encourage green retrofitting of ports with shore-power, LED yard lighting, electrified cranes. Provide:

- Interest subvention schemes for green equipment
- Tax holidays for ports achieving net-zero compliance by 2035
- Access to global climate finance via GCF and ADB under maritime-specific channels

Regional Collaboration and Policy Alignment. Solution to Regional Integration Issues (3.2 & 3.3):

India must take the lead in regional marine coordination through BIMSTEC, IORA, and IMRF by proposing:

- A "South Asia Marine Corridor Agreement" (SAMCA) to enable shared digital marine data, AI platforms, and cross-border energy logistics
- Mutual Recognition Agreements (MRAs) for port procedures and marine workforce certification
- Cross-border green corridor agreements for LNG, hydrogen, and renewable cargo between India, Bangladesh, Sri Lanka, and Maldives

AI-Driven Innovations in Marine Logistics

The evolution of artificial intelligence (AI) and digital technologies is radically transforming the landscape of marine logistics and offshore energy operations. For India and South Asia-regions historically reliant on manual processes and fragmented digital infrastructure-AI presents a unique opportunity to leapfrog into the future of smart, agile, and decarbonized marine supply chains.

This chapter explores the core AI-driven applications in marine logistics, ranging from vessel scheduling and weather routing to digital customs clearance and port automation. These innovations have already demonstrated significant operational benefits in leading maritime nations and are increasingly vital to India's ambitions under the *Maritime Amritkaal Vision 2047*.

AI in Vessel Scheduling and Berth Optimization

Inefficiencies in berth allocation and vessel turnaround are common across South Asian ports. AI-powered vessel scheduling algorithms, trained on historical port data, can:

- Predict optimal arrival windows
- Allocate berths based on cargo type, crane availability, and yard capacity
- Minimize idle time and reduce emissions from vessels at anchorage

For example, Singapore's port uses AI-driven Berth Planning Systems (BPS) that allow real-time adjustments based on vessel delays or weather disruptions. A similar system could reduce berth waiting time at major Indian ports like Jawaharlal Nehru Port (JNPT) or Paradip by 20-30%.

AI-Based Weather Routing and Navigation Optimization

Monsoon disruptions, cyclones, and swell conditions can severely impact marine operations across the Bay of Bengal and Arabian Sea. AI can be integrated into advanced weather-routing systems to:

- Predict sea conditions using satellite and historical datasets
- Adjust routes in real-time to minimize weather-related risks
- Optimize fuel consumption based on wind/current patterns

Integrating weather-AI modules with Dynamic Positioning (DP) systems on OSVs or offshore drilling vessels further enhances safety and fuel efficiency.

AI and Blockchain for Customs & Compliance Processing

Customs delays at Indian and South Asian ports add substantial time and cost to offshore logistics. An AI-enhanced customs platform can:

- Pre-screen cargo documentation
- Flag discrepancies in manifests
- Allocate digital customs officers based on predicted workload

Additionally, blockchain-backed cargo tracking systems can ensure end-to-end visibility, prevent tampering, and automatically generate compliance certificates. Ports in Rotterdam and Hamburg have piloted this successfully, achieving up to 50% reduction in customs processing time.

Predictive Maintenance in Offshore Energy Assets

Offshore oil platforms, FSRUs, and subsea pipelines are capital-intensive assets prone to wear and corrosion. Traditional inspection methods are reactive and expensive. AI-based predictive maintenance systems, paired with IoT sensors, can:

- Monitor structural stress, corrosion levels, and pressure anomalies in real time
- Alert engineers before failure occurs
- Integrate with ROV/AUV mission planning for targeted subsea inspection

For instance, AI-led maintenance of risers and conductors on platforms like Mumbai High can extend their lifecycle by 5-7 years, reduce unplanned shutdowns, and enhance HSE compliance.

AI-Enabled Port Community Systems and Data Hubs

India's Port Community System (PCS 1x) is a step toward digitization, but AI can further enhance it by:

- Providing cargo demand forecasting
- Facilitating dynamic slot booking for coastal vessels
- Suggesting alternate logistics routing during congestion

A South Asia Maritime Data Exchange (SAMDEX) platform-backed by AI-can create a regional logistics intelligence network, where ports across Sri Lanka, Bangladesh, and India share vessel traffic, weather alerts, and container flows.

AI in Offshore Renewable Energy Logistics

The future of marine operations includes offshore wind farms, floating solar platforms, and hydrogen/ammonia logistics. AI plays a key role in:

- Identifying optimal sea zones for wind turbine placement using GIS + machine learning
- Monitoring underwater cable integrity and seabed shift via AI-assisted sonar analysis
- Managing floating barge positioning and energy load balancing

Japan and Denmark are already deploying AI for these applications. India's Tamil Nadu and Gujarat coastlines offer high-potential zones where AI-enabled offshore renewables logistics can be scaled up.

Challenges in AI Adoption

While the potential is enormous, key challenges persist:

- Limited AI-readiness of public port authorities
- Inconsistent digitization across customs, DG Shipping, and port stakeholders
- Lack of marine-specific AI start-ups or R&D funding
- Cybersecurity vulnerabilities in connected port infrastructure
- Rapidly evolving technologies

The Way Forward

To overcome these, India and South Asia must:

- Launch a National AI in Marine Logistics Mission
- Fund pilot projects in AI-based scheduling, customs, and offshore IRM
- Partner with IITs, ISRO, IMU, and global marine tech firms & universities.
- Create sandbox zones within ports to test AI applications
- Incentivize Public Sector Units adoption through green compliance and ESG credits

The Taxation Bumps in Marine Logistics & Offshore Energy Infrastructure

Marine logistics and offshore energy infrastructure are heavily capital-intensive, high-risk sectors that demand robust policy and fiscal environments to function efficiently. Among the most critical levers of operational efficiency is taxation policy; a factor that often determines the competitiveness, financial viability, and investment attractiveness of a region's marine infrastructure.

While several countries across Asia and beyond have made strides in optimizing fiscal frameworks to encourage subsea and offshore investments, challenges persist across jurisdictions due to lack of harmonization, complexity in indirect taxes, and inconsistent interpretations of marine-specific exemptions. This chapter explores how taxation policies affect marine logistics operations & energy transition infrastructure, especially in relation to vessel imports, offshore support services, and fuel-related taxation, and outlines pathways to streamline and incentivize investment.

Tax Implications on Marine Equipment & Vessel Importation

The import of specialized marine equipment such as pipelay barges, offshore support vessels (OSVs), dynamic positioning units, subsea cable layers, and FSRUs is a foundational requirement for any offshore project. However, high customs duties and unclear classification of these assets under general shipping or energy project codes often result in:

- Increased upfront capital expenditure (CAPEX)

- Delayed deployment of time-sensitive offshore assets
- Complexities in temporary importation regimes (such as those used for leased vessels)

Recommendation: Introduce specific tariff codes for marine energy infrastructure and operationalize a zero-duty corridor for green maritime assets and critical subsea equipment, in line with global trade facilitation standards.

Indirect Taxation on Offshore and Subsea Services

Another friction point is the applicability of indirect taxes (like GST/VAT) on offshore services, including:

- Subsea pipeline welding
- Underwater inspections and integrity management
- EPC services for floating terminals
- Offshore crew logistics and marine catering

Because marine services often operate outside territorial waters, confusion arises over their taxability. This leads to delays in Input Tax Credit (ITC) claims and burdensome documentation for service providers.

Recommendation: Develop a "Blue-Zone Taxation Protocol", where any services rendered within the offshore energy domain (including EEZ operations) are treated under a single, well-defined tax policy-similar to special economic zones (SEZs).

Inconsistent Fuel Taxation for Green Marine Transition

The global push for decarbonization is driving adoption of alternative fuels like LNG, ammonia, methanol, and biofuels in marine logistics. However, existing fuel taxation policies often lack:

- Clear exemptions for green marine fuels
- Uniform excise duty structures across coastal states
- Infrastructure-linked tax rebates for fueling stations or bunkering depots

This results in higher operational costs and slow adoption of low-carbon fuels, especially for smaller logistics operators & port-based utilities.

Recommendation: Establish a National Green Marine Fuel Policy, with zero excise duty for certified low-carbon marine fuels, and fiscal support for setting up bunkering infrastructure at Tier-I and Tier-II ports.

Variability in Port and Vessel Taxes Across Regions

Marine logistics operators often navigate a complex matrix of:

- Port dues
- Light house fees
- Pilotage and towing charges
- Tonnage tax or harbor fees

These vary across ports and regions, creating unpredictability for planning cargo movement and offshore deployment.

Recommendation: Create a National Marine Logistics Tariff Index (NMLTI) that standardizes basic port charges & provides visibility for marine logistics planning. Introduce dynamic pricing models based on real-time berth availability, environmental performance, and cargo throughput.

Encouraging Offshore Investments Through Tax Incentives

Globally, progressive maritime economies are incentivizing investment into offshore and marine energy infrastructure through:

- Accelerated depreciation for marine assets
- Tax holidays for green port development
- R&D tax credits for marine tech innovation
- Exemptions for digital infrastructure investment (AI, blockchain, digital twins)

India & South Asia can benefit from similar policies to attract both domestic and foreign investment into:

- AI-backed port platforms
- Indigenous manufacturing of subsea systems
- Floating terminals and offshore hydrogen infrastructure

Recommendation: Launch a Maritime Innovation and Investment Incentive Scheme (MI3S) under the Ministry of Finance and MoPSW to attract greenfield investment across subsea, port-tech, and digital domains.

Tax Policy for Regional and Global Collaboration

As cross-border marine energy flows expand-particularly with floating storage units, offshore wind energy cables, and regional hydrogen corridors-there is a need for interoperable tax treaties and bilateral protocols that:

- Avoid double taxation
- Enable faster cargo movement
- Harmonize green equipment classification
- Promote mutual recognition of offshore taxation models

Recommendation: Work through regional maritime frameworks such as IORA, BIMSTEC, and IMO-led policy discussions to create a Multilateral Marine Taxation Understanding (MMTU) among participating countries.

Decarbonizing Marine Infrastructure

As the global maritime industry comes under increasing scrutiny for its carbon footprint-contributing nearly 3% of global CO₂ emissions-the imperative to decarbonize marine infrastructure has moved from aspiration to action. For India & South Asia, where offshore oil and gas activities remain vital to energy security, the challenge lies in balancing hydrocarbon-based growth with carbon-reducing innovation across ports, vessels, subsea systems, and energy logistics.

This chapter outlines the technical, strategic, and policy-driven pathways to reduce emissions across the entire marine value chain-spanning from ship design and port operations to offshore project logistics and renewable integration.

Emissions from Marine Logistics: The Baseline

Ports and marine logistics operations contribute to emissions via:

- Ship exhausts, especially from older diesel-powered OSVs & cargo vessels
- Port equipment, including RTGs, yard cranes & heavy diesel generators
- Breakwater construction, dredging, and rock-fill activities
- Offshore EPC logistics, which include helicopter transfers, anchor handling, and subsea machinery

In the absence of emission reporting standards, many emissions are underestimated or attributed to other sectors. Establishing a Port Emissions Inventory Framework (PEIF) is the first step in quantifying and addressing emissions.

Transition to Low-Carbon and Hybrid Vessels

One of the most immediate strategies is transitioning from conventional diesel engines to:

- Hybrid-electric propulsion systems
- Battery-assisted crew boats & OSVs
- Ammonia and methanol-fueled supply vessels
- LNG-powered bunker ships

For instance, Norway's Ampere and Yara Birkeland have proven that fully electric or autonomous vessels are viable for both cargo & energy operations. In India, introducing hybrid vessels for short-distance crew and cargo movement between offshore installations and ports (e.g., Mumbai High- Nhava Sheva corridor) could cut emissions by over 40%.

Policy support is needed via:

- Import duty exemptions on hybrid engine components
- Green vessel certification programs
- Interest subsidies for domestic shipyards producing green tonnage

Green Port Infrastructure

Modern ports must become energy transition hubs, featuring:

- Onshore power supply (OPS) for docked vessels to avoid idling emissions
- Electrified cargo handling equipment
- Solar rooftops & microgrids for port utilities
- Rainwater harvesting and green sewage systems

India's JNPT, Kandla, and Chennai Ports have initiated solarization and electrification drives. These should be scaled to Tier-2 & Tier-3 ports under a national "Green Port Mandate" by 2030, supported by viability gap funding (VGF).

Decarbonizing Offshore EPC Logistics

Offshore oil, gas, and wind infrastructure demand extensive logistics involving:

- Helicopter flights
- Multi-vessel coordination
- Underwater trenching and pipelaying
- New methods for reducing this footprint include:
- Digital twins to simulate and optimize vessel routes
- AI-based resource allocation
- Floating logistics bases powered by renewables

Modular breakwater construction using low-carbon materials

By integrating digital scheduling & electrified logistics at the project planning phase, EPC timelines and emissions can be minimized concurrently.

Renewable Energy Integration in Marine Projects

Decarbonization is incomplete without replacing fossil fuels with clean energy in marine-linked operations. Options include:

- Floating solar installations on port water bodies
- Wind turbines on breakwaters or coastal cliffs
- Hydrogen fuel cells for port machinery & barge power
- Wave energy devices for lighting or navigation aids

Coastal regions in Tamil Nadu, Gujarat, and Andhra Pradesh can serve as pilot locations for hybrid renewable-marine ports, with lighthouse-scale installations as proof-of-concept.

Carbon Offsets and Blue Carbon Ecosystems

Ports & coastal projects impact mangroves, coral reefs & sea-bed, which are rich in blue carbon-a natural carbon sink. Decarbonization strategy must include:

- Blue carbon restoration through mangrove plantation, seaweed farming
- Offset markets for verified marine carbon credits
- Coastal regulation zone (CRZ) adjustments to support carbon-positive development

India can position itself as a global leader in marine carbon finance, using its coastline as a nature-based infrastructure asset.

Incentivizing& Regulating Maritime Decarbonization

To accelerate the transition, regulatory and financial tools are essential:

- Green Port Index benchmarking environmental performance
- Decarbonization linked tariff discounts for cleaner ships
- Carbon intensity reporting mandates for EPC contractors
- Dedicated "Marine Carbon Transition Fund", co-financed by ports, state-owned companies & development banks

Regional cooperation through BIMSTEC or IMO-led protocols can enable harmonized standards & green financing for shared marine projects.

Human Capital & Workforce Modernization in Marine Operations

While technological innovation & policy reform are critical to the transformation of marine operations and offshore logistics, it is the human capital—engineers, technicians, seafarers, port workers, and digital specialists—that will ultimately determine the success of energy transition efforts. Yet, across India and South Asia, the marine and offshore workforce faces pressing challenges, including skill mismatches, inadequate training infrastructure, and limited exposure to emerging technologies like artificial intelligence, robotics, and digital twins.

This chapter examines the workforce landscape in the maritime and subsea domain and proposes actionable pathways to modernize marine human capital, enhance employability, and align educational ecosystems with the demands of 2040 and beyond.

The Workforce Gap: Quantitative & Qualitative Issues

South Asia, home to a significant portion of the world's seafaring and logistics personnel, paradoxically faces shortages in trained and certified professionals for modern offshore roles. Key challenges include:

- Shortfall in AI-trained port and logistics managers
- Lack of certified ROV/AUV operators and subsea technicians
- Minimal exposure to digital port infrastructure, cybersecurity, and automation
- Limited career visibility for youth in marine logistics roles beyond shipping and navy

According to a 2023 study by the Indian Maritime University (IMU), fewer than 12% of marine engineering graduates are job-ready for offshore digital projects.

Training Infrastructure: Islands of Excellence Amid Gaps

India boasts several world-class marine institutions—like the Indian Maritime University, Cochin Shipyard Training Centre, and IIT Madras Ocean Engineering—but the outreach, industry-institute linkage & digital immersion remains limited.

Key concerns:

- Most training curriculums are not aligned with real-time offshore project demands
- Industry internships, particularly in FSRUs, offshore wind, or deep-sea logistics, are rare
- Limited online training platforms for continuous learning, particularly in coastal and rural zones

Emerging Skillsets for the Energy Transition Era

The future of marine operations demands a hybrid skillset that blends:

- Marine engineering + Data analytics
- Subsea operations + Remote vehicle handling
- Project logistics + Digital twin simulation
- Port management + AI-based scheduling + customs automation

A comprehensive Marine Skills Roadmap 2047 must focus on equipping professionals across four levels:

- Frontline field operators- trained in AI-supported inspection and digital customs tools
- Technicians and engineers- skilled in hybrid vessel systems and offshore wind assembly
- Mid-level managers- equipped with project analytics, ESG reporting, and IRM planning
- Strategic planners- conversant with carbon trading, hydrogen corridors, and global policy frameworks

Opportunities for India: Global Marine Talent Hub

India, with its large young population and English proficiency, is well positioned to emerge as a global exporter of skilled marine professionals. A forward-looking human capital strategy should include:

- Export-ready training certifications aligned with IMO, ABS, DNV standards
- Bilateral partnerships with countries investing in offshore renewables (e.g., Japan, Norway, UAE)
- Virtual reality (VR) & simulation-based academies for offshore EPC and subsea inspection
- National Skill Development Corporation (NSDC) collaboration with marine companies & private ports

Industry-Academia-Government Collaboration

To close the skills gap sustainably, the marine sector needs:

- Joint curriculum development involving IITs, maritime universities, and offshore contractors
- Onboard training mandates for OSVs, tugs, and pipelayers through internships
- Government-backed "Blue Collar to Blue Ocean" schemes, encouraging workers in coastal zones to transition to high-paying offshore roles

Incentivize skill-building through:

- Skill-linked tax credits for marine companies training young professionals
- Subsidies for coastal digital training centers equipped with subsea and port-tech labs
- Establishing a Maritime Centre of Excellence for AI and decarbonization

Gender and Inclusivity in the Marine Workforce

The future marine workforce must also be diverse & inclusive. Current participation of women in port & offshore logistics remains below 8%, and there are significant gaps in representation from marginalized and tribal coastal communities.

Solutions include:

- Gender-inclusive policies & safe offshore infrastructure
- Scholarships and mentorship for women in marine STEM programs
- Inclusion of traditional coastal knowledge systems into formal training (e.g., navigation, meteorology, tide prediction)

Skill Accreditation and Mobility

India should advocate for a South Asian Maritime Skills Agreement, under which certifications are mutually recognized across BIMSTEC & IORA regions. This would enable faster mobilization of skilled personnel during offshore emergencies & cross-border energy project deployments.

Create a Digital Marine Skills Passport (DMSP)-a secure, blockchain-based platform that stores qualifications, simulator results & sea-time logs-enhancing credibility and mobility of professionals across borders.

Breakwater Construction, Offshore Ports & Floating Infrastructure

As maritime economies evolve to meet the dual imperatives of energy security and environmental sustainability, the importance of adaptive, low-carbon, and resilient port and offshore infrastructure has come into sharp focus. With rising sea levels, increased cyclone frequency, and heightened ESG (Environmental, Social, and Governance) accountability, India and South Asia must rethink how breakwaters, ports, terminals, and floating infrastructure are planned and constructed.

This chapter details the emerging engineering practices, green materials, and energy-integrated infrastructure shaping the next generation of marine assets, with particular focus on gravity-based offshore terminals, eco-sensitive jetty design, and tidal power integration.

Gravity-Based Offshore Storage Terminals (GBST)

A Gravity-Based Offshore Storage Terminal (GBST) is a fixed, subsea or seabed-sitting structure that stores liquefied energy products-like LNG, green ammonia, or hydrogen-near offshore production or transport corridors. These terminals are constructed using pre-cast caissons or reinforced concrete tanks & are positioned using their own weight (gravity), requiring no pile-driving or seabed penetration.

Advantages of GBST:

- Environmentally non-intrusive: No major seabed drilling or pile foundations
- Earthquake and cyclone resilience due to low centre of gravity
- Faster deployment compared to land-based or floating terminals
- Expandable through modular caisson additions
- Lower operating risk than floating terminals in high swell zones

Use Cases for India and South Asia:

- Along the Gulf of Khambhat, where high tidal velocity limits floating options
- For offshore ammonia storage before conversion to hydrogen at shore
- As part of strategic marine energy reserves, isolated from onshore urban constraints

Policy Recommendation: GBSTs should be included in India's "Blue Economy Infrastructure Vision 2047" and supported through green bond financing and marine EPC fast-track clearances.

Eco-Sensitive Jetty & Pier Construction

Traditional jetty construction, especially in ecologically fragile coastal areas like Sundarbans, Lakshadweep, or Gulf of Mannar, often leads to:

- Habitat disruption (coral reefs, mangroves)
- Altered tidal flows and sediment transport
- Thermal discharge concentration zones

To mitigate these, new methods have emerged:

a) Piled Deck Jetties with Low Impact Footprints

Instead of solid fill or sheet-piled jetties, piled deck structures elevate the jetty over sensitive seabed, allowing marine life and tidal flows to continue unhindered below. Modern jetty piles can be coated with biofilm-friendly epoxy to promote reef colonization.

b) Pre-fabricated Modular Jetty Segments

Segments are assembled on land and launched by barge, avoiding on-site underwater work. This minimizes sediment disturbance, underwater noise pollution, and marine waste.

c) Jetty-Zone GIS Planning

Using geospatial modelling and drone-based bathymetry, jetty placement is optimized to avoid biodiversity hotspots, seagrass meadows, or breeding areas of endangered species.

Recommendation: All future marine infrastructure in sensitive zones should undergo Ecological Impact Simulation (EIS) using digital twins & be certified under an "Eco-Jetty Seal" by the Ministry of Environment.

Tidal Power Plants: A Decarbonization Imperative

India, Bangladesh, and parts of Southeast Asia possess strong tidal amplitudes, especially in estuarine regions like:

- Gulf of Khambhat (tidal range > 11 meters)
- Sundarbans delta
- Mekong and Irrawaddy estuaries

Despite this, tidal power remains underutilized due to initial capex concerns. But global benchmarks from South Korea's Sihwa Lake, France's La Rance, and UK's Swansea Bay show how tidal plants can deliver stable, long-duration energy-ideal for port, logistics & shore power.

Advantages:

- Predictable & non-intermittent energy
- Minimal land requirement
- Zero carbon during operation
- Can be co-located with fish nurseries and sediment retention structures

Integration with Marine Infrastructure:

- Ports powered by tidal turbines embedded in breakwaters or causeways
- Tidal energy used to power offshore hydrogen electrolyzers
- Use of tidal barrage gates as silt control for river-fed ports like Kolkata

Action Plan:

- Launch a Tidal Infrastructure Mission (TIM) under MNRE
- Establish 3 pilot projects by 2030 with port co-location mandates
- Allow private EPC consortia to invest via VGF (Viability Gap Funding) model

Modular Floating Logistics Platforms

Floating platforms-either semi-submersible, pontoon-based, or tethered barges-are now increasingly used to create temporary or permanent logistics nodes for:

- Offshore wind component assembly
- FSRU-based LNG transfer
- Transshipment in deep-sea zones
- Marine waste recovery or desalination units
- These platforms:
 - Require minimal seabed work
 - Can be relocated based on project demand
 - Use hybrid power systems (solar, wave, wind)
 - Are scalable and replicable for island logistics or disaster relief

Example: The Netherlands' Port of Rotterdam is undertaking trials of floating docks for hydrogen refueling barges. India could emulate this in Kandla, Vizhinjam, or Paradip.

Breakwater Innovations for Climate Resilience

Traditional rock-mound breakwaters face failure risks due to sea-level rise, increased wave energy, and erosion. Future-ready breakwaters include:

- Hybrid breakwaters with dynamic core materials that absorb and adapt to swell
- Wave energy capturing breakwaters, which integrate tidal turbines within the wall
- Bio-engineered breakwaters, supporting mangrove or oyster reef growth
- Floating breakwaters linked to AI-based sensors that adjust buoyancy and position

In high-risk cyclone zones like Odisha & Bangladesh coasts, these can prevent multi-million-dollar offshore damages.

Renewables in Marine Logistics & Project Development

The maritime industry-long synonymous with fossil-fueled logistics and offshore oil & gas-is undergoing a profound transformation. As the world transitions toward cleaner energy systems, renewables are now redefining marine operations, not only in terms of powering ships and port infrastructure but also as core components of offshore project development. Wind, solar, tidal, and hydrogen-based technologies are increasingly finding their place within the design and execution of marine logistics chains.

This chapter explores how renewable energy is being embedded across the lifecycle of marine infrastructure, from shore-to-ship power systems & green bunkering to offshore wind farms and hybrid energy corridors. It also analyses policy drivers, technological innovations, & regional collaborations that can accelerate India and South Asia's leadership in renewable marine integration.

Offshore Wind Projects: The New Energy Frontier

Globally, offshore wind is emerging as one of the most promising renewable energy frontiers. Asia is leading the charge, with countries like China, Vietnam, and Japan making large investments. India has a target of 30 GW offshore wind capacity by 2030, particularly in Gujarat and Tamil Nadu.

Marine logistics requirements for offshore wind:

- Transport of monopiles, blades, nacelles from coastal manufacturing hubs

- Use of jack-up vessels and cable-laying ships
- Installation of floating substations
- Undersea cabling to shore grids

India's Gulf of Khambhat & Rameshwaram coast offer optimal conditions. These projects will require the expansion of green ports capable of handling oversized renewable cargo and specialized vessels.

Solarization of Ports & Floating Solar Projects

India's port ecosystem is gradually embracing solarization as a decarbonization and cost-saving measure. Ports like JNPT, Vishakhapatnam, and Tuticorin have initiated rooftop and yard solar installations.

Emerging practices include:

- Floating solar PV on port basins and reservoirs
- Solar-powered cold storage and warehousing
- Solar-powered drone stations and surveillance systems
- Electrified port cranes and forklifts supported by on-site solar microgrids

Best Practice: Singapore's Jurong Port derives over 60% of its operational electricity from solar-India could replicate this model at Paradip, Mormugao & Haldia ports.

Hydrogen as a Marine Fuel and Logistics Enabler

Hydrogen, particularly green hydrogen derived from renewables, is emerging as a key enabler of marine energy transition.

Applications in marine logistics:

- Fuel cells for tugboats, ferries, and port equipment
- Hydrogen bunkering terminals integrated into ports
- Hydrogen caverns and gravity-based floating terminals near the shore
- Hydrogen-powered autonomous inspection vessels and drones

Indian initiatives: The Ministry of Ports, Shipping & Waterways (MoPSW) has identified Kandla and Paradip as potential sites for hydrogen hubs, supported by the National Hydrogen Mission.

Future Vision: Develop "Hydrogen Ready Ports" with integrated bunkering, electrolysis stations powered by tidal/wind, and hydrogen export terminals for global markets.

Biofuels & Green Ammonia Integration

Ammonia and bio-LNG are gaining traction as zero-carbon shipping fuels, particularly for deep-sea logistics. Ports must now accommodate:

- Ammonia imports and storage terminals
- Green ammonia pipelines to industrial clusters
- Compatible bunkering systems
- Ammonia cracking systems for hydrogen conversion

South Asian collaboration is key. India, Japan & UAE are exploring joint investments in ammonia transport corridors, with pilot projects along the west coast.

Integration of Renewable Energy in Marine EPC Projects

Large-scale offshore projects-from subsea pipelines to FSRU terminals-consume high levels of power during construction.

Renewables can help as below:

- Power remote construction camps through solar + battery hybrids
- Use floating wind turbines to energize offshore EPC logistics hubs
- Provide clean power to digital equipment, reducing genset dependence
- Enable green commissioning of marine assets

EPC contractors like L&T Hydrocarbon, Sapura, and McDermott are increasingly adopting modular clean energy systems on barges and jack-up platforms.

Marine Energy Corridors: Regional Connectivity Through Renewables

Future marine logistics will be shaped by renewable-based energy corridors. These are integrated systems linking:

- Offshore renewable generation (wind, wave, solar)
- Subsea transmission (HVDC cables)
- Hydrogen/ammonia logistics terminals
- Green ports with digital customs and AI-driven energy trading

India's Role: Act as a hub between Gulf producers & Southeast Asian consumers, offering green refueling, renewable assembly, and smart grid connectivity via coastal ports.

Strategic Concept: The Bay of Bengal Blue Energy Corridor (BBEC)-a BIMSTEC-driven regional marine energy logistics chain that includes Bangladesh, Sri Lanka, Thailand, and Myanmar.

Policy Support & Financing Tools

To realize these renewable-marine synergies, policy interventions must focus on:

- Green port incentives under *Sagarmala & Maritime Amritkaal Vision 2047*
- Priority clearances for floating solar and offshore wind zones
- Carbon credit trading mechanisms linked to marine renewables
- Multilateral financing through ADB, JICA, World Bank for floating and green energy terminals
- Dedicated Renewable Marine EPC Clusters at coastal industrial corridors

Policy Alignment & Regional Cooperation for Sustainable Marine Logistics

The marine logistics & offshore energy sectors are inherently transnational-dependent on global shipping routes, shared ocean resources, cross-border energy corridors, and integrated regulatory mechanisms. To ensure long-term sustainability, resilience, and innovation, there is a pressing need for harmonized policies & strategic cooperation across South Asia, Southeast Asia, and beyond.

India's maritime future, particularly under its Maritime *Amritkaal* Vision 2047, must be rooted in collaborative frameworks that align with international best practices while also addressing regional nuances in regulation, climate adaptation, digitalization, and workforce mobility.

This chapter explores policy-level imperatives and regional cooperation models that can accelerate the shift toward a green, efficient, and future-proof marine logistics ecosystem.

The Case for Regional Marine Policy Alignment

Differences in marine regulation across South and Southeast Asian countries-ranging from port digitization standards to environmental codes & taxation rules-create bottlenecks for shipping lines, EPC contractors, and offshore operators. Key challenges include:

- Non-uniform adoption of IMO environmental protocols
- Varying customs clearance systems and port handling norms
- Disparate certifications for subsea technicians and offshore crew
- Inconsistent carbon emission tracking and offset standards
- To address these, countries in the region must explore maritime regulatory harmonization, especially in:
 - Subsea asset inspections
 - Crew safety and evacuation protocols
 - Fuel standards (ammonia, LNG, hydrogen)
 - Offshore digital tracking and green tax incentives

India's Maritime Amritkaal Vision 2047: A Policy Anchor

India's Maritime *Amritkaal* Vision 2047, launched by the Ministry of Ports, Shipping & Waterways, aims to position the country as a global maritime leader through:

- Development of 500 plus MTPA port capacity
- Modernization of shipbuilding & ship repair yards
- Greening of ports & development of hydrogen/ammonia refueling terminals
- Increased coastal shipping share from 6% to 15% of domestic cargo
- Establishment of digital port ecosystems integrated with blockchain, AI, and IoT

This vision aligns with global goals like IMO 2030 and 2050 GHG strategies and offers a foundational policy structure to integrate regional cooperation efforts.

Cross-Border Green Corridors & Infrastructure Protocols

India, Sri Lanka, Bangladesh, Thailand & ASEAN nations can collaborate on marine green corridors, supported by:

- Common carbon-neutral targets
- Joint investment in tidal/wind-powered terminals
- Unified ESG reporting standards
- Shared offshore hydrogen/ammonia logistics pipelines

Pilot Project Suggestion: The Bay of Bengal Maritime Energy Corridor, connecting green hydrogen hubs across India (Odisha), Bangladesh (Chittagong) & Sri Lanka (Trincomalee), supported by a common customs blockchain for cargo traceability.

Digital Port Protocols & AI-Driven Customs Harmonization

Today, over 60% of delays in cross-border marine logistics stem from asymmetric digital maturity between ports. India's Port Community System (PCS 1x) has made progress, but integration with neighboring countries is still nascent.

Solutions:

- Establish a South Asian Marine Digitalization Charter (SAMDIC)
- Develop common APIs for AI-based scheduling, cargo tracking, weather routing
- Joint digital skills training for port staff and customs officials

Enable real-time vessel data exchange and shared port slot visibility

Joint Marine Risk and Emergency Response Frameworks

Given increasing climate risks, countries need unified protocols for:

- Cyclone alerts and port shutdown guidelines
- Shared emergency response vessels and equipment caches
- Coordinated offshore evacuation drills
- Harmonized risk rating and insurance assessment models for marine EPC projects

These can be formalized under a South Asia Marine Safety and Resilience Pact, supported by UN bodies or regional forums like BIMSTEC and IORA.

Regional Marine Talent Mobility and Credential Recognition

To support future offshore infrastructure and renewables, the region needs skilled workforce mobility. A platform for mutual recognition of offshore certifications, such as ROV operator licenses, diving technician training, and green port management degrees, will:

- Reduce project staffing delays
- Promote collaboration in joint ventures
- Allow marine companies EPC firms to pool resources across borders

Proposal: Launch a "Marine Skills Passport"-a digitally verified credential system for port and subsea workers, linked to skill councils and accreditation boards.

Financing Sustainable Marine Projects through Regional Banks

Green and blue economy financing will play a critical role. Collaboration with:

- Asian Development Bank (ADB) for marine renewable EPC
- AIIB and JICA for subsea cable infrastructure & green ports
- GCF (Green Climate Fund) for decarbonization-linked port upgrades
- Cross-border blue bonds issued jointly by partner countries

This will reduce risk premiums and crowd in private capital for floating terminals, tidal power, and hydrogen corridors.

India and Indian Companies in the Marine and Port Sector

India, with its vast coastline of over 7,500 kilometers, 12 major ports, and more than 200 non-major ports, holds a strategic geographical position in the Indo-Pacific maritime trade network. As the world undergoes an energy transition and embraces digital and decarbonized logistics models, Indian marine and port infrastructure is undergoing unprecedented transformation, with both public sector undertakings (PSUs) and private firms contributing significantly to capacity building, offshore energy expansion, and sustainable port modernization.

This chapter surveys key developments, investments, and future strategies across India's marine and offshore sector, including port infrastructure, subsea engineering, offshore terminals, digital shipping, and green fuel logistics. It also outlines India's role in the South Asian maritime future and how Indian companies are emerging as global marine logistics partners.

India's Port Infrastructure and Capacity Expansion

India has made significant strides in improving port performance, capacity utilization, and turnaround time. Some key achievements include:

- Increase in total cargo handling capacity to over 2,600 MTPA (million tons per annum)
- Average turnaround time at major ports reduced to <2.5 days (vs. 4.5 days in 2014)
- Electrification and shore power installations in ports like JNPT, Paradip, and Deendayal Port (Kandla)
- Automation of container handling, RFID-based cargo tracking, and PCS 1x digital ecosystem

The government's flagship *Sagarmala* Programme, with a total outlay of INR 6 lakh crore, includes more than 800 projects focused on:

- Port modernization
- Connectivity enhancement
- Port-linked industrialization
- Coastal community development

Indian Public Sector Units Leadership: HPCL, IOCL, GAIL, and ONGC

India's oil and gas PSUs are actively venturing into marine infrastructure and logistics:

- HPCL (Hindustan Petroleum Corporation Limited)
 - Owns & operates LNG terminal at Chhara.
 - Partner in HMEL's Bhatinda project, with port-based crude import strategies
 - Exploring gravity-based storage units for offshore green fuel imports
 - Evaluating FSRU-based LNG and ammonia handling capabilities for future non-fuel diversification
 - Can be a potential developer of hydrogen bunkering ports and hydrogen cavern clusters on the coasts.

- IOCL (Indian Oil Corporation Limited)
 - Operating Kamarajar LNG Terminal & planning expansion in Dhamra, Ennore
 - Commissioning India's first green hydrogen microgrid at its port-linked refinery
 - Integrating digital twin modeling for port planning and marine fuel logistics
- GAIL (India) Ltd.
 - Developing subsea pipeline systems to connect offshore gas fields
 - Investing in ammonia import facilities to feed its planned hydrogen infrastructure
 - Tied up with global marine AI firms to enable real-time cargo dispatch planning
- ONGC
 - India's largest offshore operator, with mature fields in the Mumbai High and KG Basin
 - Operating multiple jack-up rigs, FPSOs, and pipeline-laying vessels
 - Exploring deep-water and ultra-deep-water hydrocarbon basins using advanced subsea engineering
 - Partnering in offshore wind studies in Gujarat

Private Sector Participation in Marine and Offshore Development

Indian private companies are becoming global players in marine EPC and port logistics:

- Adani Ports & SEZ (APSEZ)
 - India's largest port operator, with control over 30% of national cargo volume
 - Operates deep-draft ports at Mundra, Dhamra, Vizhinjam, and Hazira
 - Building India's first transshipment hub at Vizhinjam with green shipping lanes
 - Developing green ammonia terminal infrastructure at Dhamra
- L&T Hydrocarbon Engineering
 - Leading EPC contractor for subsea pipelines, FPSO integration, and modular offshore terminals
 - Delivered projects in Saudi Arabia, Qatar, and Indonesia-positioning India as an offshore engineering export hub
 - Investing in floating hydrogen terminals and gravity-based ammonia storage modules
- Reliance Industries Limited (RIL)
 - Operating deep-water gas fields via Reliance-BP JV
 - Building ammonia and green hydrogen terminals in Jamnagar SEZ
 - Exploring AI-based marine scheduling systems for logistics efficiency

Subsea and Offshore Equipment Manufacturing

India is emerging as a manufacturing base for subsea and marine energy systems:

- Tata Steel, Welspun & Ratnamani produce offshore-grade steel pipes and risers

- Bharat Heavy Electricals Ltd. (BHEL) manufactures marine turbines, cable-laying modules
- Mazagon Dock Shipbuilders and Cochin Shipyard now design and export DP2-class OSVs and inspection vessels
- The Kattupalli port cluster hosts fabrication yards for wind jacket structures and subsea cable storage units

India's Blue Economy and Vision 2047

The *Maritime Amritkaal* Vision 2047 sets ambitious goals:

- Transform India into a top-5 maritime nation
- Develop 100% green energy-powered ports
- Expand India's offshore renewable energy capacity (wind, hydrogen)
- Create hydrogen-ready shipping lanes and subsea cable hubs
- Enhance marine digital infrastructure to global standards

India's Blue Economy is expected to contribute \$1 trillion to GDP by 2047, with a marine logistics & energy backbone that supports:

- Floating infrastructure deployment
- Marine AI and IoT adoption
- Hydrogen and ammonia trade corridors
- Export of marine EPC and digital services

Conclusion and Strategic Roadmap- Anchoring India at the Global Maritime Crossroads

As the world moves into a decade marked by climate urgency, geopolitical shifts, and energy diversification, marine operations and logistics are undergoing a historic transformation. Offshore oil, gas, and renewables are no longer a silo regional endeavors but they are deeply interconnected through transcontinental energy flows, supply chain partnerships, and environmental policies.

In this changing landscape, India occupies a uniquely central role, both geographically and strategically. Surrounded by three major water bodies—the Arabian Sea, Bay of Bengal, and the Indian Ocean, India serves as the bridge between Middle Eastern producers, Southeast Asian industrial centers, and Western consumer markets. Its maritime future is no longer just about cargo but it is also about connectivity, energy, and sustainability.

The Global Offshore Investment Landscape

In 2024, the global offshore energy investment surged across regions, as shown below:

Table 1

Region	Offshore Oil & Gas Investment 2024 (USD Billion)
North America	130
Middle East	120
Asia-Pacific	112

Region	Offshore Oil & Gas Investment 2024 (USD Billion)
Europe	98
Africa	60
Russia & CIS	45

This investment reflects global momentum toward securing energy resilience amid decarbonization & geopolitical uncertainty. India's strategic goal must be to embed itself more deeply in this investment matrix—not just as a market or port of call, but as a marine energy and technology hub.

India's Strategic Position in Global Energy Corridors

India is already the world's third-largest energy consumer. The vast majority of its crude oil and LNG imports pass through marine corridors. The future offers new opportunities:

- Acting as a floating terminal base for green ammonia & LNG storage for regional refueling
- Becoming the hydrogen export hub to Japan, Korea, and Europe, using offshore caverns & subsea pipelines
- Offering EPCI and subsea engineering services to neighboring countries via Indian state-owned companies & private contractors
- Leading joint energy corridors with GCC nations, ASEAN, and Africa using shared marine infrastructure

India's marine future must be designed not just for throughput but for thought-leadership.

Alignment with Global Energy Strategies

India's goals converge well with the ambitions of other major energy players:

Table 2

Country/Region	Marine/Offshore Focus	Synergy with India
Middle East (UAE, Saudi Arabia)	Hydrogen, ammonia shipping; FSRUs	Joint hydrogen bunkering & floating terminals
Europe (Norway, Netherlands)	Offshore wind, green ports	Port automation & offshore EPC best practices
Asia-Pacific (Japan, South Korea)	Hydrogen imports, ship design	Export of green hydrogen and floating storage
Russia & CIS	Subsea gas pipelines, Arctic ports	Subseainspection, monitoring partnerships

These synergies highlight how Indian ports, shipyards and state-owned energy companies can build regional alliances in hydrogen, renewables, digital marine systems, and green shipping.

Future Priorities for Indian Marine Strategy

To future-proof its marine economy, India must focus on:

A. Energy Transition Readiness

- Build gravity-based terminals for ammonia/hydrogen
- Expand tidal and floating solar capacity

- Construct green-certified breakwaters & floating ports
- B. Subsea Engineering and Safety
 - Train 25,000+ subsea workers under marine skill passport
 - Deploy autonomous subsea drones & ROVs for IRM
 - Strengthen EPRS (Emergency Pipeline Repair Systems)
- C. Digital Marine Ecosystem
 - AI-based multi-port scheduling and customs visibility
 - Real-time voyage planning and carbon tracking
 - Integration with international PCS systems
- D. Global Partnerships
 - Establish South Asia Green Marine Alliance (SAGMA)
 - Joint R&D with Norway, Japan, UAE on hydrogen and tidal storage
 - Lead Maritime G20 Track for blue economy financing

Roadmap for Indian Public & Private Stakeholders

Table 3

Stakeholder	Strategic Imperative
HPCL, IOCL, BPCL	Develop FSRUs, hydrogen caverns, floating terminals
ONGC, GAIL	Invest in deep-water drilling, subsea AI systems
ADANI, L&T, RIL	Lead in gravity-based storage, green EPC exports
MoPSW	Implement Maritime <i>Amritkaal</i> Vision 2047 fully
SHIPPING MINISTRY	Harmonize marine tax and digital infrastructure

Final Word: Oceans as Engines of Energy Transition

The 21st century's marine economy is about more than ports & tankers. It is about:

- Oceans as energy collectors (wind, tidal, hydrogen)
- Ports as clean energy platforms
- Ships as floating AI-enabled assets
- Marine infrastructure as climate resilience architecture

Asia, with its human capital, shipbuilding expertise, coastline, and policy ambition, is uniquely placed to lead the global marine renaissance-offering not just throughput, but thought, not just logistics, but leadership.

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Appendix: Acronyms & Full Forms

Acronyms & Full Forms

ADB:	Asian Development Bank
AI:	Artificial Intelligence
APSEZ:	Adani Ports & Special Economic Zone
BBEC:	Bay of Bengal Blue Energy Corridor
BPS:	Berth Planning Systems
CAPEX:	Capital Expenditure
CRZ:	Coastal Regulation Zone
DMSP:	Digital Marine Skills Passport
DNV:	Det Norske Veritas (Classification Society)
DP:	Dynamic Positioning
EIS:	Ecological Impact Simulation
EPRS:	Emergency Pipeline Repair Systems
ESG:	Environmental, Social, and Governance
FDI:	Foreign Direct Investment
GAIL:	Gas Authority of India Limited
GBST:	Gravity-Based Offshore Storage Terminal
GCF:	Green Climate Fund
HPCL:	Hindustan Petroleum Corporation Limited
IMU:	Indian Maritime University
IOCL:	Indian Oil Corporation Limited
ITC:	Input Tax Credit
JNPT:	Jawaharlal Nehru Port Trust
MMTU:	Multilateral Marine Taxation Understanding
MoPSW:	Ministry of Ports, Shipping & Waterways
MTPA:	Million Tonnes Per Annum
NMLCA:	National Marine Logistics Coordination Authority
NMLTI:	National Marine Logistics Tariff Index
NMSDP:	National Marine Skill Development Program
NSDC:	National Skill Development Corporation
OPS:	Onshore Power Supply
PDMI:	Port Digital Maturity Index
PSA:	Port of Singapore Authority
RIL:	Reliance Industries Limited
SAMDEX:	South Asia Maritime Data Exchange
SAGMA:	South Asia Green Marine Alliance
SAMDIC:	South Asian Marine Digitalization Charter
TIM:	Tidal Infrastructure Mission

UNCTAD:	United Nations Conference on Trade and Development
UNEP:	United Nations Environment Programme
VGF:	Viability Gap Funding
VR:	Virtual Reality